

Deep Learning-Based Coconut Ripeness Classification and Yield Estimation using YOLO and Kalman Filtering

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Abstract—Accurate detection and classification of coconuts according to their ripeness stages are essential for optimizing agricultural harvesting practices, reducing manual labor, and increasing economic efficiency. The ripeness of coconuts significantly influences their commercial value, nutritional content, and suitability for various products such as coconut water, coconut milk, and oil extraction. However, manual inspection for ripeness determination is labor-intensive, subjective, and prone to inaccuracies. To address these challenges, this paper proposes a deep learning-based methodology employing YOLOv11 integrated with the Kalman Filter-Hungarian Algorithm, YOLOv8, and a custom-designed Convolutional Neural Network (CNN) to accurately identify coconuts at three distinct ripeness stages: immature, mid-ripe, and fully ripe. An open-source coconut dataset was utilized and enhanced through data augmentation techniques, including rotation, scaling, and color adjustments, to improve the robustness and generalization capabilities of the models across varying environmental and lighting conditions. The models were trained and rigorously evaluated using standard object detection metrics, specifically mean Average Precision (mAP50 and mAP50:95). Furthermore, the proposed approach was extended to facilitate automated counting of coconuts using different tracking methodologies, demonstrating practical applicability and potential integration into real-time agricultural monitoring systems. Experimental results illustrate that the proposed deep learning models effectively differentiate coconut ripeness stages, providing a significant advancement for precision agriculture and automation in coconut farming.

Index Terms—Deep Learning, Object Detection, YOLOv8, Kalman Filter, Coconut Ripeness Classification

I. INTRODUCTION

Accurate yield estimation in agriculture is crucial, as it allows farmers to optimize resource allocation,

improve harvest logistics, and enhance market planning. Traditional methods for estimating yield, which typically rely on manual sampling and destructive measurements, are labor-intensive, time-consuming, and prone to significant error. In recent years, advances in computer vision (CV) have enabled the development of non-destructive, real-time yield estimation systems that offer greater accuracy [1]. Early approaches to fruit yield estimation primarily used handcrafted features—such as color, texture, and shape—to segment fruit from the background. For example, color thresholding techniques were applied to canopy images to detect apples and grapes. However, these methods often struggled under varying illumination conditions, limiting their practical use in real-world environments [2].

The advent of deep learning has significantly transformed the field of automated yield estimation. Convolutional Neural Networks (CNNs) have shown remarkable ability to learn discriminative features directly from raw images, overcoming many of the limitations associated with traditional methods. For instance, Bargoti and Underwood [3] successfully employed CNN-based segmentation to achieve high-precision fruit detection in orchards, while Gongal et al. [4] combined stereo vision with deep learning to enhance apple crop-load estimation. These studies demonstrate that robust feature learning and multi-view imaging can substantially improve detection accuracy, even under challenging conditions such as occlusions and variable lighting [5].

Despite these advances, many existing systems remain constrained by their reliance on controlled imaging environments and often require extensive post-processing to correct for issues like double-counting in sequential frames. Additionally, most current methods focus on estimating fruit

counts without fully addressing spatial distribution, which is critical for mapping yield variability across large fields [6]. A comprehensive yield mapping system should not only count fruit but also integrate geospatial information to generate high-resolution yield maps. Such maps are invaluable for site-specific management and resource optimization [7].

In this paper, we aim to address these challenges by developing an integrated system that combines deep learning for fruit detection with robust tracking algorithms to ensure that each fruit is counted only once. Our approach facilitates rapid decision-making and allows farmers to adjust management practices in real time, ultimately leading to improved crop performance and reduced operational costs.

II. RELATED WORK

Recent advances in fruit detection and maturity estimation have increasingly shifted from traditional image processing techniques to deep learning-based approaches. Early machine vision systems relied on manually engineered features such as color, shape, and texture for fruit segmentation and classification, but were often limited by variations in illumination and background complexity [2]. To overcome these limitations, researchers have increasingly turned to deep learning methods.

YOLO-based models have been particularly influential in advancing fruit ripeness detection. Xiao et al. [8] introduced a YOLOv8-based framework for fruit ripeness identification, demonstrating improved accuracy and robustness across varying environmental conditions. Similarly, Yang et al. [5] enhanced YOLOv8 by integrating it with a lightweight Transformer module (LW-Swin Transformer), significantly improving strawberry ripeness detection under occlusions and partial visibility.

Lightweight adaptations of YOLO architectures have also been proposed for specific agricultural applications. Chen et al. [9] developed YOLOv8-CML, a streamlined model that improves the detection of color-changing melon ripeness while reducing computational complexity, making it suitable for real-time agricultural monitoring. Likewise, Zhang et al. [10] proposed an improved YOLOv8-based maturity recognition algorithm for Xinhui citrus, incorporating multiscale feature fusion to boost precision under complex field conditions.

Beyond vision-only approaches, Ouhda et al. [2] explored combining YOLO detection with K-means clustering for maturity analysis of date fruits, achieving higher accuracy compared to purely vision-based methods. Additionally, Wang et al. [7] utilized UAV-based imagery combined with deep learning to detect litchi fruit maturity, addressing challenges associated with wide-area agricultural monitoring.

Convolutional Neural Networks (CNNs) have also played a major role in fruit classification. Han and Yi [11] proposed a deep CNN model specifically for palm fruit maturity classification, achieving strong generalization performance even with smaller datasets. Similarly, Caladcad and Piedad [11] and Sattar [12] explored acoustic signal-based deep learning approaches for coconut maturity estimation, offering

non-visual methods that complement traditional imaging techniques.

Ma et al. [1] provided a comprehensive review of state-of-the-art deep learning techniques for fruit maturity detection, outlining challenges such as dataset imbalance, model overfitting, and real-time deployment issues. Additional studies by Li et al. [6] and Khan et al. [13] have focused on fruit yield mapping and palm fruit ripeness detection, respectively, using deep learning models that integrate both visual and spatial information.

Other recent innovations include Nguyen and Tran's [14] Fruit-Net system, a CNN-based multi-fruit recognition framework, and Abdul Rahman et al.'s [15] intelligent system for predicting orange fruit ripeness, combining image processing and deep neural networks for better harvesting decision support.

Building upon these advancements, our approach leverages YOLOv8 architectures combined with robust tracking algorithms like the Kalman filter and the Hungarian algorithm to achieve accurate fruit detection, ripeness classification, and yield estimation in dynamic agricultural environments.

III. DATA COLLECTION AND PREPROCESSING

The methodology adopted for detecting and classifying coconuts across different ripeness stages (Stage 1, Stage 2, and Stage 3) follows a systematic pipeline comprising data collection, preprocessing, model training, and performance evaluation. The dataset, sourced from an open-source repository, was annotated with bounding boxes to indicate coconuts and their respective ripeness stages. Rigorous preprocessing was conducted, including data cleaning,

Algorithm 1 Coconut Detection and Classification Methodology

Require: Open-source dataset of annotated coconut images

Ensure: Trained models and performance metrics

1: **Data Collection:**

2: Obtain dataset annotated with bounding boxes for different ripeness stages (immature, mid-ripe, fully ripe)

3: **Data Preprocessing:**

4: Remove duplicates and low-quality images

5: Perform data augmentation: rotation, flipping, scaling, mosaic, Gaussian noise, cutout

6: Split dataset into 70% training, 15% validation, and 15% testing sets

7: **Model Training:**

8: Train YOLOv11 integrated with Kalman Filter-Hungarian Algorithm

9: Train YOLOv8 using pre-trained COCO weights

10: Train custom CNN with dropout regularization

11: **Model Evaluation:**

12: Evaluate models on test set using precision, recall, mAP₅₀, and mAP_{50:95}

13: Select the best-performing model based on validation metrics

augmentation (such as rotation, flipping, scaling, and mosaic augmentation), and splitting into training, validation, and testing subsets as shown in Algorithm 1 [15].

Three models—YOLOv11 integrated with the Kalman Filter–Hungarian Algorithm, a custom Convolutional Neural Network (CNN), and YOLOv8—were trained using optimized configurations tailored to each architecture. These configurations incorporated appropriate hyperparameters, optimizers, and loss functions to ensure robust detection and classification. GPU-accelerated training enabled efficient experimentation, while standard object detection metrics, including precision, recall, mAP_{50} , and $mAP_{50:95}$, were used to evaluate model performance on the testing set. The methodology ensures reproducibility and scalability by leveraging widely used frameworks and open-source resources [2].

A. Data Collection

The dataset utilized in this study was obtained from an open-source repository, ensuring both accessibility and reproducibility. It consists of high-resolution images of coconuts captured in their natural environment, annotated with bounding boxes indicating their location and ripeness stage (Stage 1, Stage 2, and Stage 3). The annotations followed standard formats compatible with object detection frameworks such as YOLO [13].

B. Preprocessing and Augmentation

To enhance the flexibility and generalizability of the models, comprehensive preprocessing and data augmentation techniques were applied to the raw dataset. These steps are critical for addressing challenges such as class imbalance, variations in illumination, and environmental noise [16]. The preprocessing steps included:

1) *Data Cleaning*: Low-quality or duplicate images, which could negatively impact model training, were removed. Additionally, annotation accuracy was carefully verified to ensure consistency throughout the dataset [14].

2) *Data Augmentation*: Augmentation techniques were applied to the original images to expand the dataset and introduce variability, thereby improving the model's ability to generalize to unseen scenarios. Figure 1(a) illustrates the original image and various augmented samples. The following augmentation techniques were employed: Images were rotated at random angles to simulate different viewpoints as shown in Fig. 1(e). Horizontal and vertical flips were performed to account for orientation changes as shown in Fig. 1(f). Blurring was applied to a subset of images to simulate sensor noise and varying focus conditions as shown in Fig. 1(c). In Fig. 1(b) and Fig. 1(d), random noise was added to simulate real-world sensor imperfections, and four different images were combined into a single image to expose the model to varied object sizes and contexts. Random regions of images were masked to encourage the model to learn from incomplete object patterns [10].

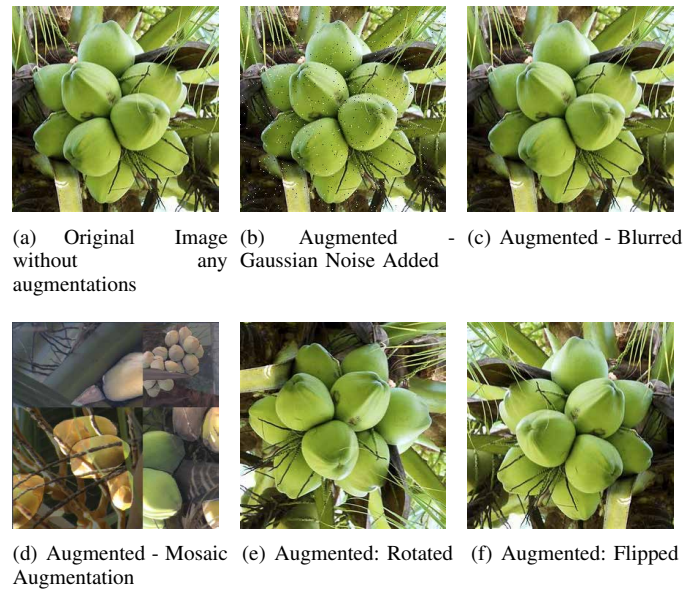


Fig. 1. Original and Various Augmented Images: (a) Original, (b) Gaussian Noise, (c) Blurred, (d) Mosaic Augmentation, (e) Rotated, and (f) Flipped

3) *Data Splitting*: Following augmentation, the dataset was divided into three subsets: 70% is used for learning model parameters, 15% for monitoring model performance during training and preventing overfitting, and 15% for evaluating the final performance of the trained models [7].

C. Annotation Format

Annotations were stored in YOLO format, where each bounding box is represented by the class label, normalised centre coordinates, and normalized width and height. This format ensures compatibility with all object detection models employed in this study. The combination of open-source data, rigorous preprocessing, and diverse augmentation techniques provided a strong foundation for model training and evaluation, ensuring the developed models' applicability to real-world scenarios.

IV. TRAINING METHODS

In this section, the training methodologies employed for object detection and ripeness classification of coconuts are presented. Three distinct models were developed and evaluated: YOLOv11 integrated with a Kalman filter and the Hungarian algorithm, a custom-designed Convolutional Neural Network (CNN), and YOLOv8 utilizing transfer learning. Each model was trained with tailored configurations concerning optimization strategies, learning schedules, loss functions, and regularization techniques to maximize detection accuracy and generalization performance. The following subsections detail the specific training setups, hyperparameter choices, and enhancements incorporated for each model.

A. Custom Convolutional Neural Network (CNN)

A custom Convolutional Neural Network (CNN) architecture was developed, comprising convolutional

and pooling layers, followed by fully connected layers for final classification. To mitigate the risk of overfitting, dropout layers were incorporated, and hyperparameters were tuned through iterative experimentation. The training procedure utilized the Adam optimizer with an initial learning rate of 0.0001, a batch size of 32 images per batch to expedite training while preserving generalization capabilities, and 50 epochs with early stopping based on the validation loss plateau. The cross-entropy loss function was employed for ripeness classification tasks. While the custom CNN achieved high training accuracy, it exhibited signs of overfitting during validation, attributed primarily to limitations in the dataset size and diversity [17].

B. YOLOv8

The YOLOv8 model, recognized for its efficiency in object detection tasks, was trained using transfer learning from pre-trained weights on the COCO dataset. Training employed the Adam optimizer alongside a dynamic learning rate scheduler to adjust the learning rate in response to validation performance. A batch size of 16 images per batch was selected to maintain consistent gradient updates. The model was trained for 25 epochs, employing both standard and advanced data augmentation techniques, including mosaic and mixup, to enhance generalization. For bounding box regression, the Complete Intersection over Union (CIoU) loss was utilized, while focal loss was adopted for classification tasks. The YOLOv8 model exhibited rapid convergence and demonstrated robust performance across all ripeness stages of coconuts. Throughout the training process, all models were systematically monitored, and the best-performing checkpoints were selected based on validation metrics for subsequent evaluation.

C. YOLOv11 with Kalman Filter and Hungarian Algorithm

The YOLOv11 model, integrated with a Kalman filter and the Hungarian algorithm, was trained to enhance object detection accuracy and tracking stability. The training configuration employed Stochastic Gradient Descent (SGD) as the optimizer, with an initial learning rate of 0.001 and a momentum coefficient of 0.9. A batch size of 16 images per batch was utilized to balance memory consumption and training stability. The model was trained for 100 epochs, with checkpoints implemented to preserve the best-performing weights based on validation loss. A composite loss function comprising localization, confidence, and classification components was adopted, specifically optimized for the detection of coconuts at various stages of ripeness. The incorporation of the Kalman filter contributed to improved temporal consistency in tracking, while the Hungarian algorithm facilitated effective association of detections across frames [17].

D. Explanation of Metrics

Table I presents the key performance metrics used to evaluate the models' effectiveness in detecting coconuts at

various stages of ripeness. A detailed explanation of each metric is provided below.

1) *Precision (%)*: Precision is defined as the ratio of correctly identified positive instances (true positives) to all instances predicted as positive (true positives and false positives):

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \quad (1)$$

2) *Recall (%)*: Recall, also known as sensitivity, is the ratio of correctly identified positive instances to all actual positive instances:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (2)$$

3) *Mean Average Precision at IoU = 0.5 ($mAP_{50}\%$)*: The mAP_{50} metric measures the average precision of detections where the Intersection over Union (IoU) between predicted and ground truth bounding boxes is at least 0.5.

4) *Mean Average Precision at IoU = 0.5:0.95 ($mAP_{50:95}\%$)*: This stricter metric averages the precision across multiple IoU thresholds ranging from 0.5 to 0.95, in steps of 0.05.

V. RESULTS AND DISCUSSION

To assess the effectiveness of various approaches in detecting and classifying coconuts at different stages of ripeness (Stage 1, Stage 2, and Stage 3), three deep learning-based models were evaluated: YOLOv11 with Kalman Filter and Hungarian Algorithm, a custom Convolutional Neural Network (CNN), and YOLOv8. The results obtained from these models are summarized in the following subsections.

YOLOv11 was integrated with a Kalman Filter and the Hungarian Algorithm to enhance tracking stability and detection performance. This model achieved a moderate overall mean Average Precision (mAP_{50}) of 56.9% and a $mAP_{50:95}$ of 43%. The inference speed was efficient, requiring only 5.0ms per image, thus rendering the model suitable for real-time applications despite its relatively lower precision in detecting Stage 1 coconuts. A custom CNN was designed with dropout layers and learning rate adjustments to mitigate overfitting. The model achieved 100% accuracy on the validation set, with precision, recall, and F1-score all

TABLE I
PERFORMANCE COMPARISON OF THE TESTED MODELS FOR COCONUT RIPENESS DETECTION.

Model	Stage	Precision (%)	Recall (%)	mAP_{50} (%)	$mAP_{50:95}$ (%)
YOLOv11 with KF-HA	Stage 1	36.7	54.1	32.8	23.7
	Stage 2	53.4	81.0	75.6	59.7
	Stage 3	65.3	70.1	62.4	45.5
Custom CNN	Stage 1	100.0	100.0	–	–
	Stage 2	100.0	100.0	–	–
	Stage 3	100.0	100.0	–	–
YOLOv8	Stage 1	44.4	49.6	38.8	27.9
	Stage 2	57.9	78.9	74.3	58.0
	Stage 3	70.9	67.0	68.8	47.8

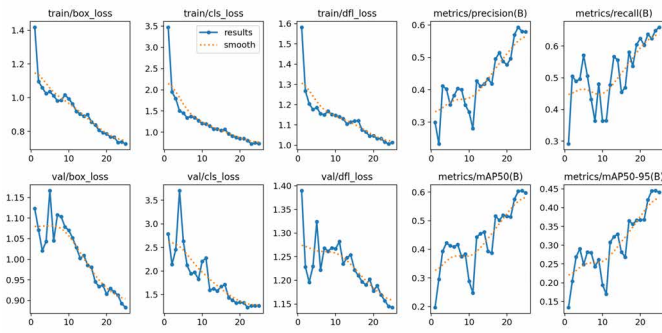


Fig. 2. Loss and precision metrics from YOLOv8

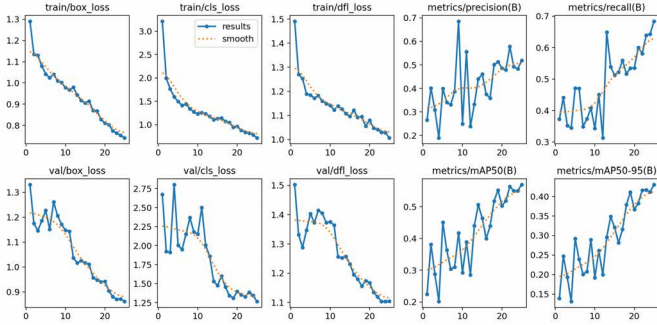


Fig. 3. Loss and precision metrics from YOLOv11 with KF-HA

attaining 1.00. However, such perfect performance strongly suggests overfitting, likely due to limited dataset diversity and insufficient complexity to challenge the model during training. YOLOv8, trained over 25 epochs, demonstrated the best overall performance across all evaluation metrics. It achieved a mAP₅₀ of 60.6% and a mAP₅₀₋₉₅ of 44.6%. Although marginally slower than YOLOv11, YOLOv8 exhibited more balanced performance across all ripeness stages and showed greater robustness, particularly in detecting Stage 2 and Stage 3 coconuts, as illustrated in Table I.

A. Comparative Analysis

Among the evaluated models, YOLOv8 offered the most favorable trade-off between precision, recall, and mAP metrics, excelling particularly in Stage 2 and Stage 3 classifications. In contrast, the custom CNN, although achieving perfect scores, exhibited signs of overfitting, attributable to the limited variability of the training data. YOLOv11, while demonstrating competitive inference speeds, struggled with lower precision in Stage 1 detection. Overall, the results indicate that YOLOv8 is the most suitable model for the given task. Nevertheless, further dataset expansion and augmentation are recommended to improve detection accuracy, particularly for Stage 1 coconuts. Comparative results are depicted in Fig. 2, and Fig. 3.

Figures 2 and 3 present the training loss and precision metrics for YOLOv8 and YOLOv11 with Kalman Filter–Hungarian Algorithm (KF-HA), respectively. YOLOv8 (Fig. 2) shows a smooth decline in loss and consistent

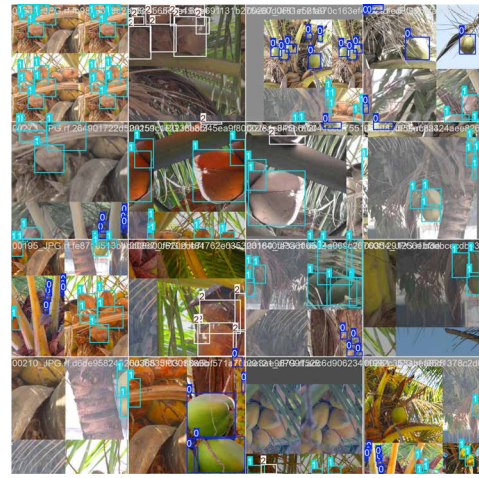


Fig. 4. Sample Results Images- YOLOv11 with KF-HA



Fig. 5. Sample Results Images- YOLOv8

increase in precision, indicating stable convergence and strong generalization. In contrast, YOLOv11 (Fig. 3) exhibits fluctuating validation loss, suggesting unstable learning and sensitivity to noise. While YOLOv11 benefits from improved tracking, it performs less reliably in early-stage detection. Overall, YOLOv8 demonstrates better training behavior and is more suited for real-time coconut ripeness classification.

B. Counting Techniques

After identifying and classifying coconuts with our detection models, we integrate counting methods to estimate yield in dynamic scenes. Two complementary approaches are employed: a frame-based counting method and a line-crossing technique. In this straightforward approach, the total number of detections in each frame is summed to estimate the number of coconuts present as shown in Fig. 4. Although computationally efficient, this method can overestimate counts when the same object appears in multiple consecutive frames. To overcome the limitations of simple frame-based counting, we implement a line-crossing method. A virtual line is defined

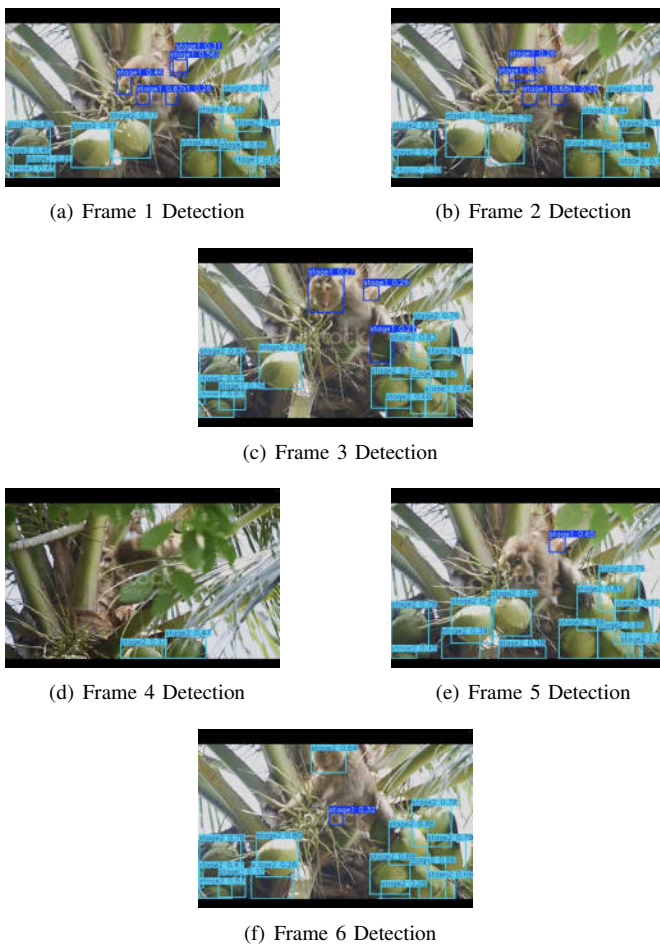


Fig. 6. Video Detection Across Frames: (a) Frame 1, (b) Frame 2, (c) Frame 3, (d) Frame 4, (e) Frame 5, and (f) Frame 6

within the region of interest, and each coconut is tracked across frames using a Kalman filter. The Hungarian algorithm is applied for robust data association between detections over time. A coconut is counted only once, at the moment its trajectory intersects the predefined line. This approach reduces duplicate counts and improves accuracy, particularly in scenarios with overlapping objects or camera motion as shown in Fig. 5. Figure 6 illustrates the detection using YOLOV11 and KF-HA.

VI. CONCLUSION

In this study, a deep learning-based framework for coconut ripeness classification and yield estimation was proposed, leveraging the strengths of YOLOv11 with Kalman Filtering and the Hungarian Algorithm, a custom Convolutional Neural Network (CNN), and YOLOv8. By systematically applying data augmentation techniques and optimizing model architectures, the proposed models demonstrated robust performance under varying environmental conditions. Among the evaluated models, YOLOv8 achieved the highest accuracy and generalization capabilities, making it the most suitable candidate for real-world deployment. Furthermore,

the integration of tracking algorithms significantly improved counting accuracy, reducing duplication errors across frames. This work provides a practical solution for automating coconut harvesting and yield estimation, which can enhance efficiency in agricultural operations. Future work will focus on expanding the dataset, incorporating geospatial mapping for yield distribution analysis, and exploring lightweight models for real-time deployment on mobile and edge devices.

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